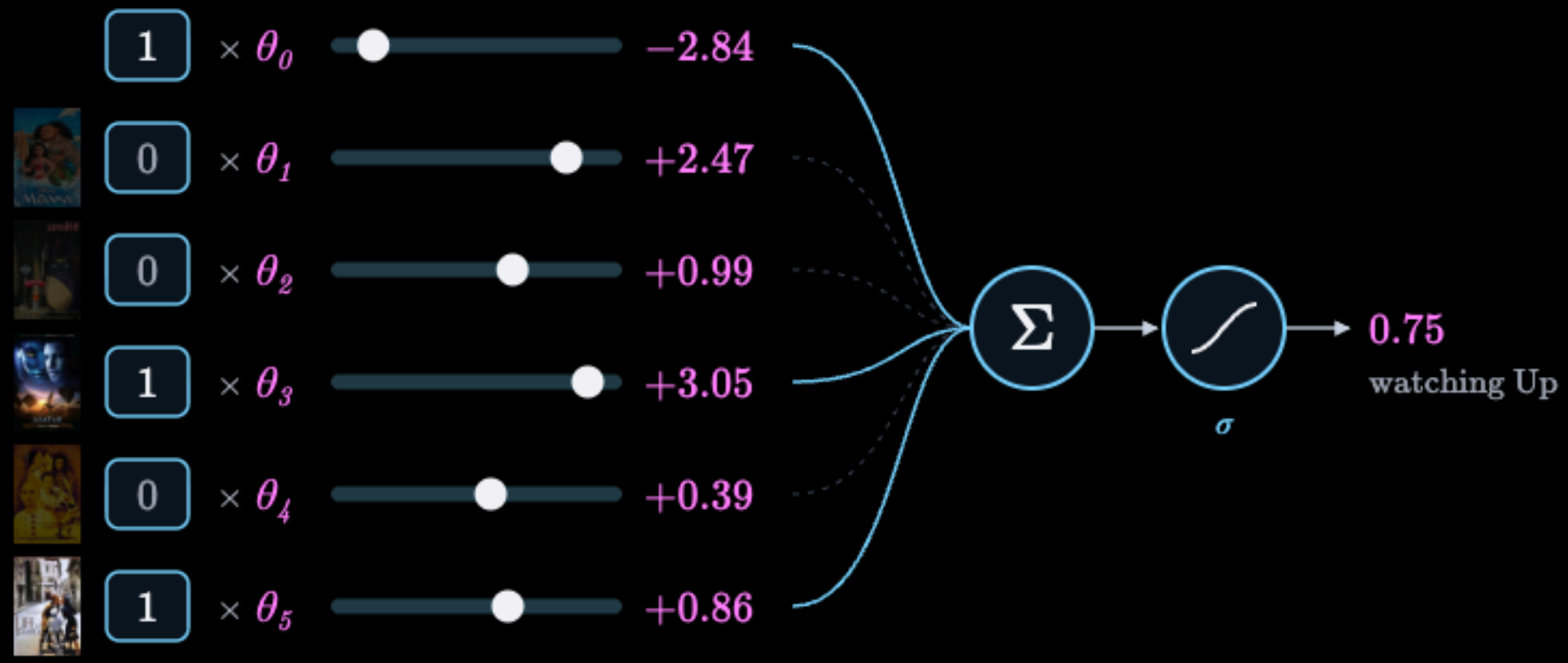
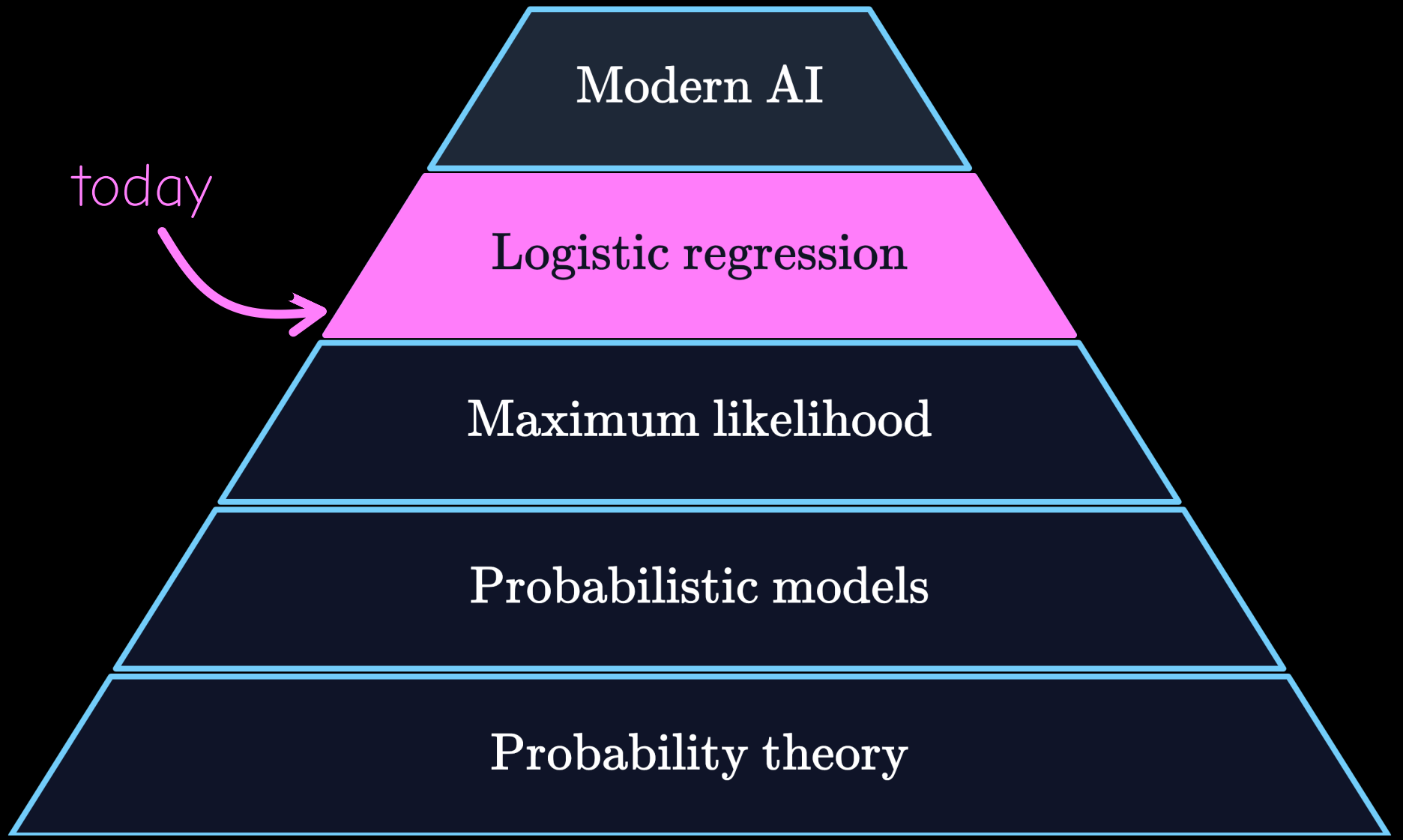

Logistic Regression



A single artificial neuron

Logistic regression, the model we build today.

The big picture



Learning goals

1. Recognize the logistic regression probabilistic model

2. Train it with a computer, by climbing the log likelihood



[Background]


LELAND STANFORD JUNIOR UNIVERSITY
DIE LUFT DER FREIHEIT WEHT
1891

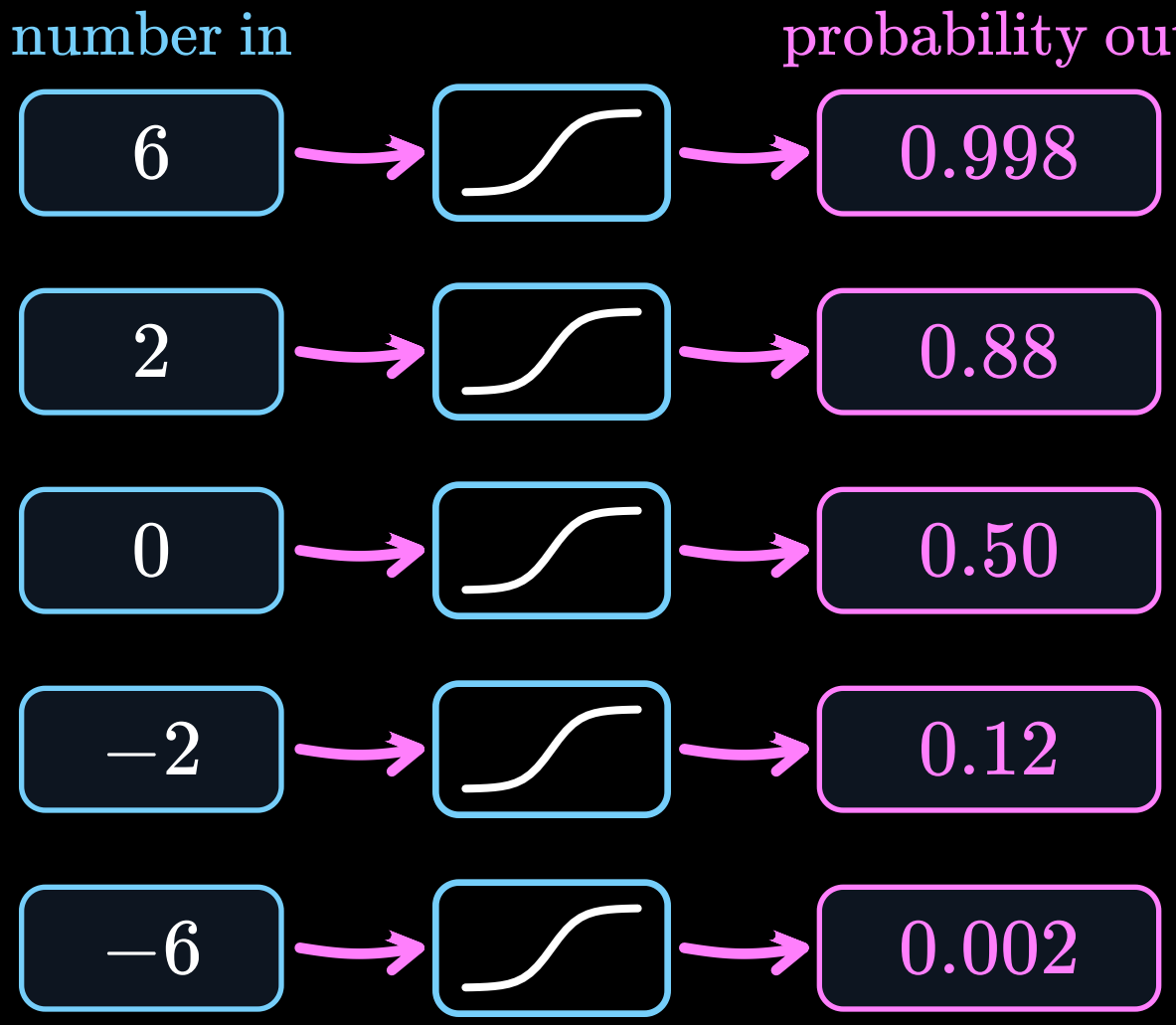
The sigmoid function

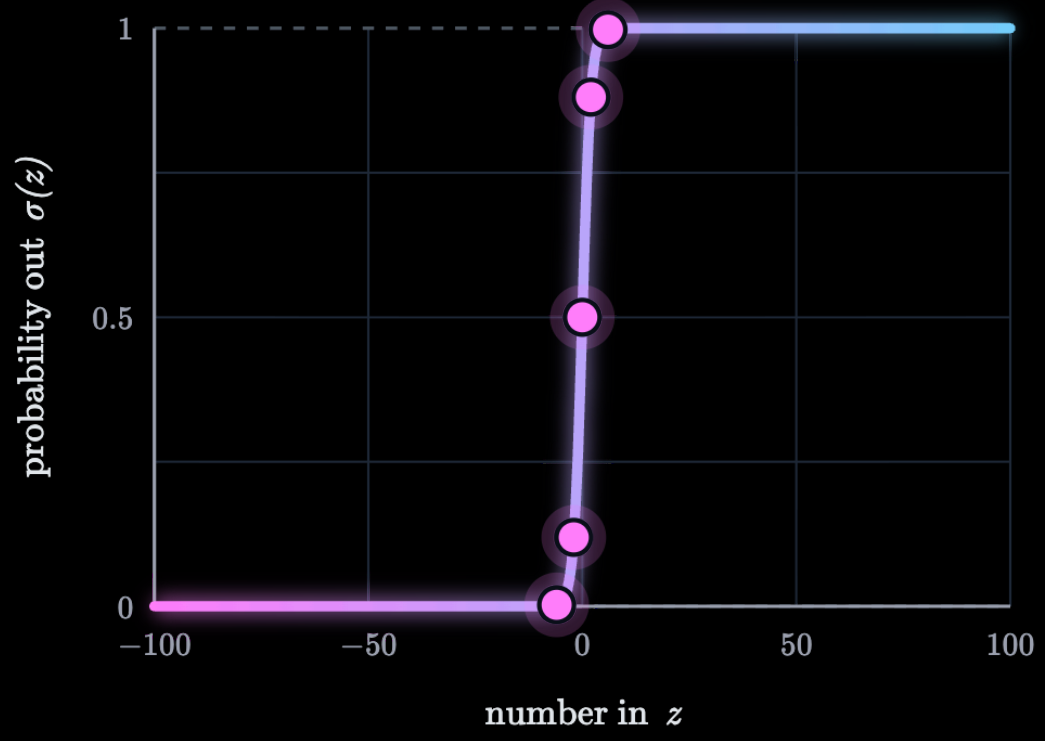


always between 0 and 1



The graph of the sigmoid function

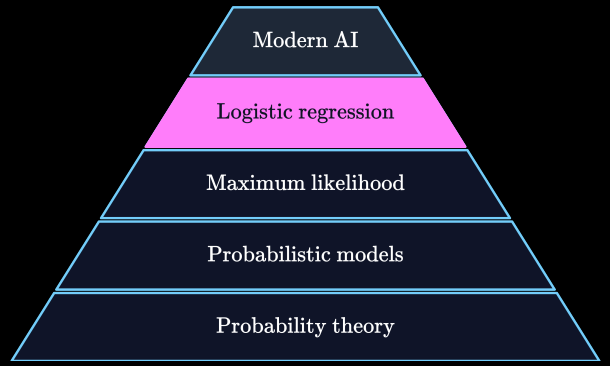




$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

[End Background]





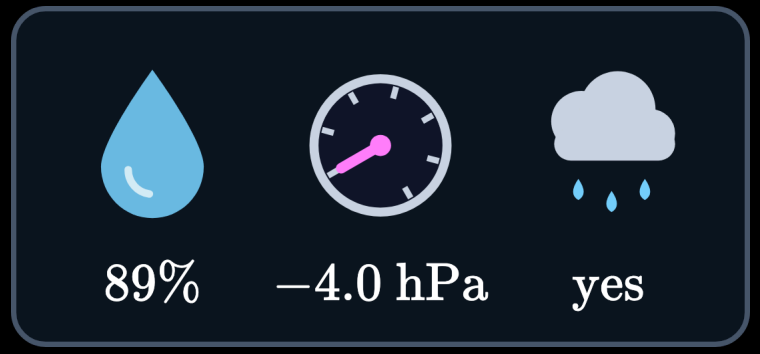
Today we build one of the artificial neurons we saw inside GPT-2.

What problems can one neuron solve?

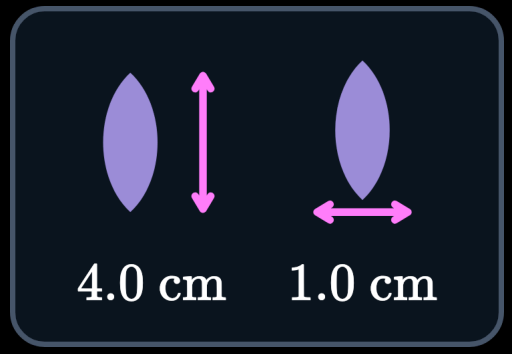
Three (seemingly) different problems



Movie recommendation



Rain forecasting



Flower identification

features

what they watched



probabilistic model



probability watched Up

	feature 1	feature 2	feature 3	feature 4	feature 5	label
						
viewer	Moana	My Neighbor Totoro	Avatar	Crouching Tiger, Hidden Dragon	Life Is Beautiful	watched Up?
1	0	0	1	0	1	yes
2	0	0	0	0	0	no
3	1	0	1	0	0	yes
4	0	1	0	0	0	no
5	0	0	0	1	1	no

features
today's weather



probabilistic model



probability of rain

	feature 1	feature 2	feature 3	label
				
day	humidity	fall, hPa	rained today	rain tomorrow?
Nov 1, 2010	89%	−4.0	yes	yes
Nov 2, 2010	79%	−11.9	yes	no
Nov 3, 2010	53%	10.2	no	no
Nov 4, 2010	66%	5.1	no	yes
Nov 5, 2010	90%	5.4	yes	yes



features

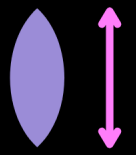
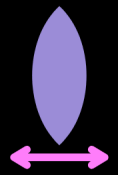
the petal's size



probabilistic model

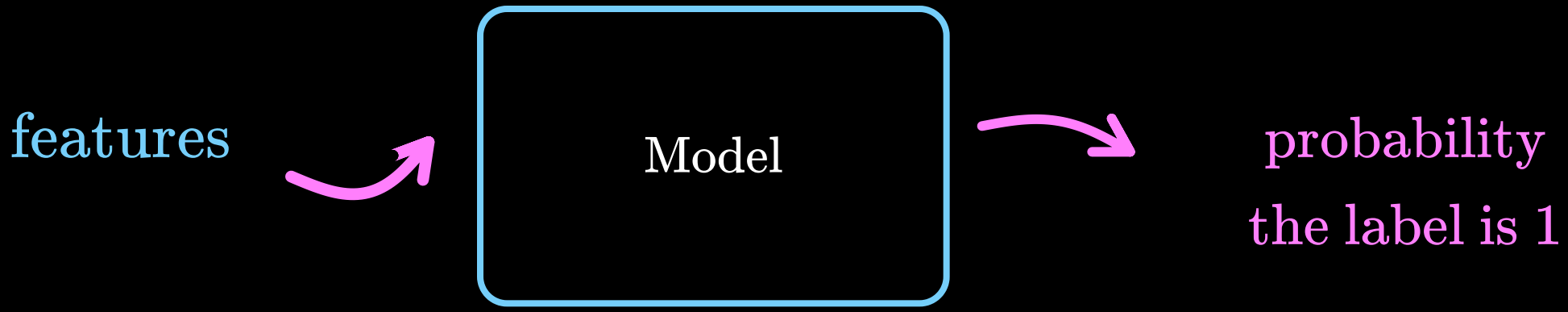


probability virginica

flower	feature 1  petal length	feature 2  petal width	label  virginica?
1	4.0 cm	1.0 cm	no
2	4.4 cm	1.2 cm	no
3	6.0 cm	2.5 cm	yes
4	5.6 cm	2.4 cm	yes
5	5.7 cm	2.3 cm	yes



The classification task





Example 1

Movie recommendation

The movie data

	x_1	x_2	x_3	x_4	x_5	y
						
viewer 1	0	0	1	0	1	1
viewer 2	0	0	0	0	0	0
viewer 3	1	0	1	0	0	1
viewer 4	0	1	0	0	0	0
viewer 5	0	0	0	1	1	0
viewer 6	0	1	1	1	0	1

x: what we know

y: watched Up?

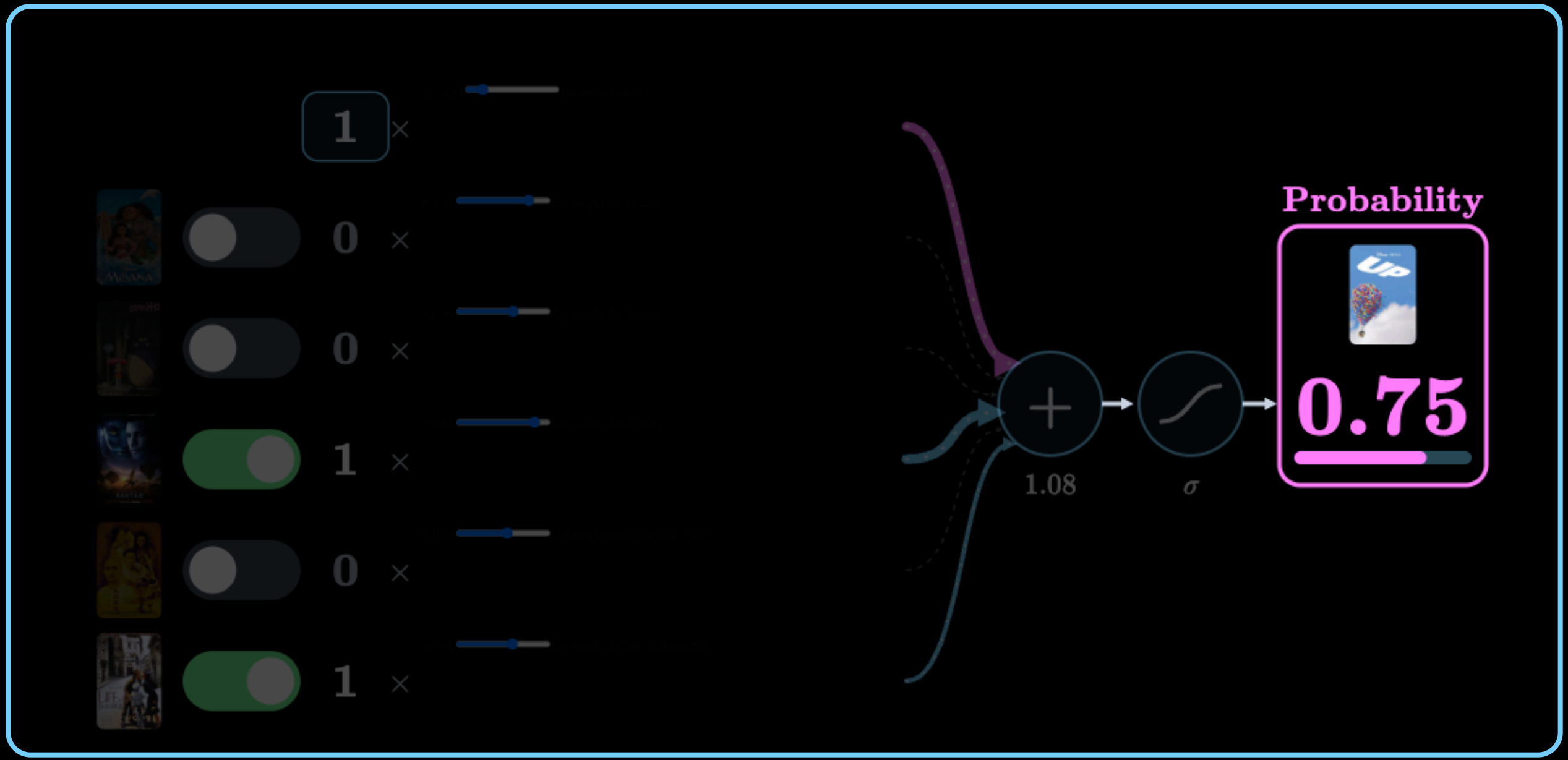
One feature

	x_1	x_2	x_3	x_4	x_5	y
						
viewer 1	0	0	1	0	1	1
viewer 2	0	0	0	0	0	0
viewer 3	1	0	1	0	0	1
viewer 4	0	1	0	0	0	0
viewer 5	0	0	0	1	1	0
viewer 6	0	1	1	1	0	1

x_3: watched Avatar?

Classification

Logistic regression



The parameters



one weight for every input

What kind of model is this?





Logistic regression is a probabilistic model.

The output is the probability of a Bernoulli!

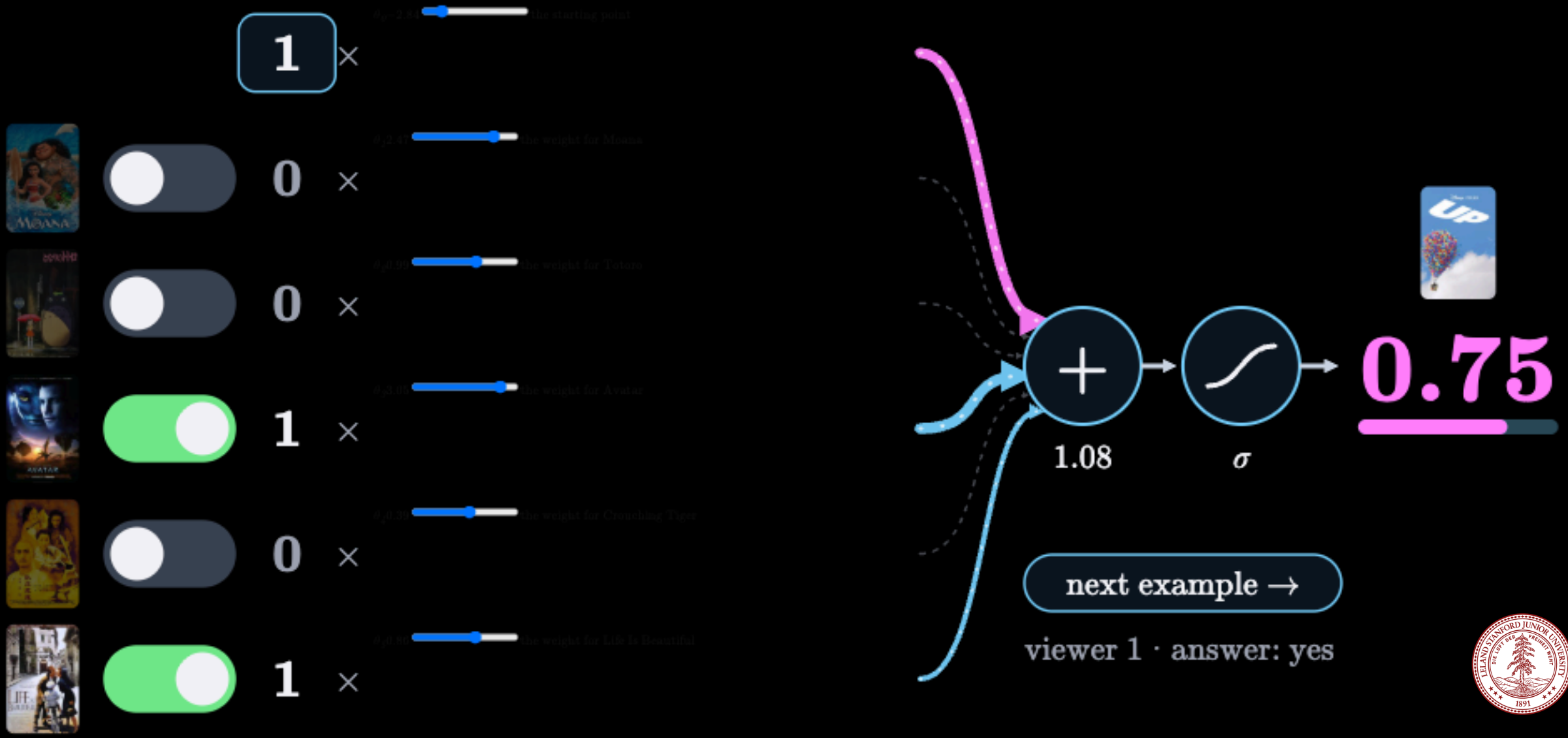




All models are wrong, some models are useful.

Logistic regression doesn't describe how the world works. It is, however, a good predictor.

The Logistic Regression Model

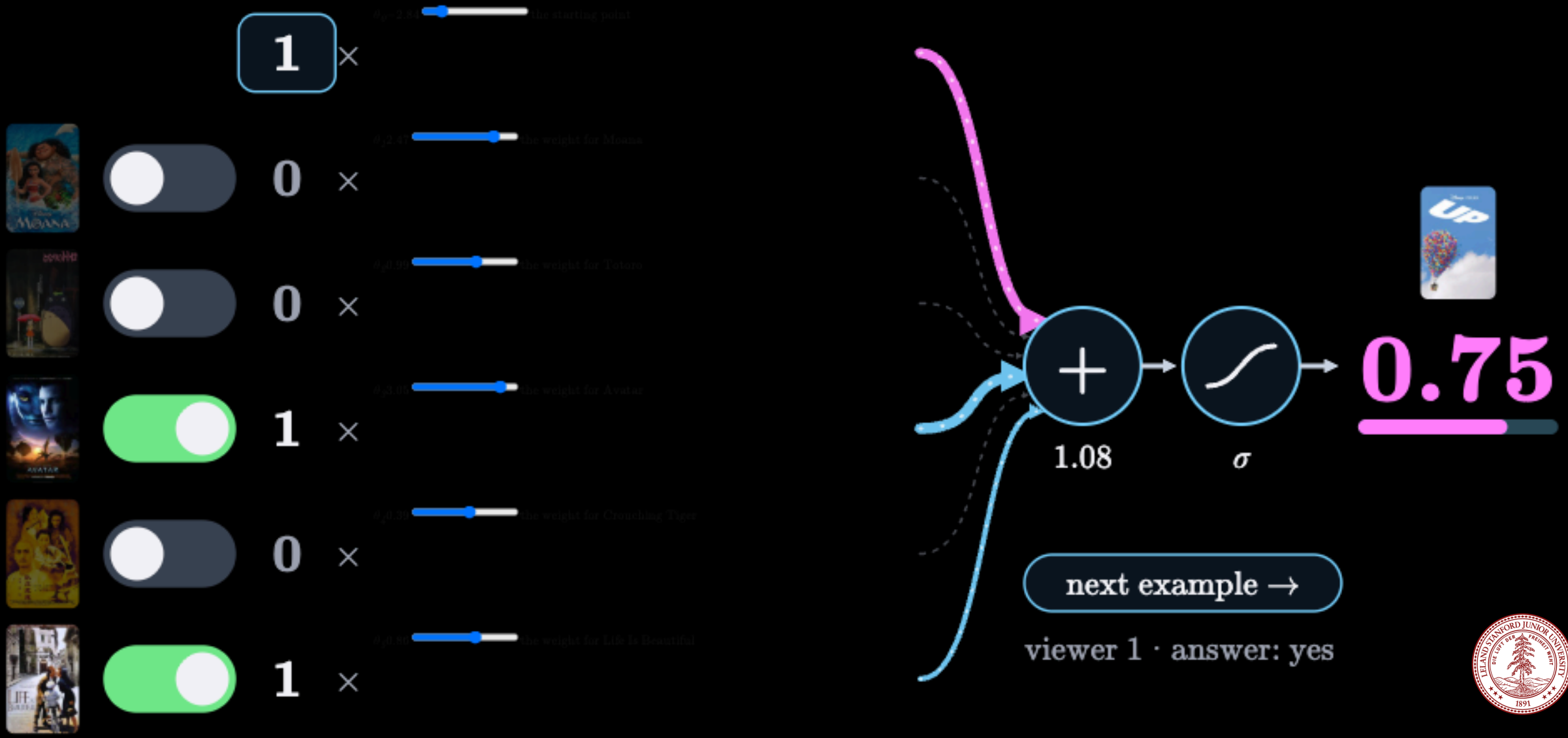


What is the difference
between a “smart” model and
a “dumb” model?

The parameters!



The Logistic Regression Model



The model for one viewer

What they watched



0 0 1 0 1

Their chance of Up 0.75 predicted by the model

The parameters: one θ for each movie, plus the bias θ_0.

$$\mathrm{P}(\text{yes} \mid x) = \sigma(\theta_0 + \theta_1 x_1 + \cdots + \theta_m x_m)$$



We wrote logistic regression as a probabilistic model with six parameters.

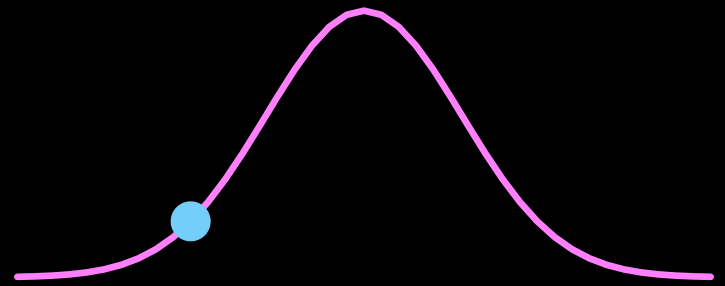
How do we choose the six θs?



The MLE recipe

1. Write the model ✓
2. Score a guess
3. Find the slope
4. Climb

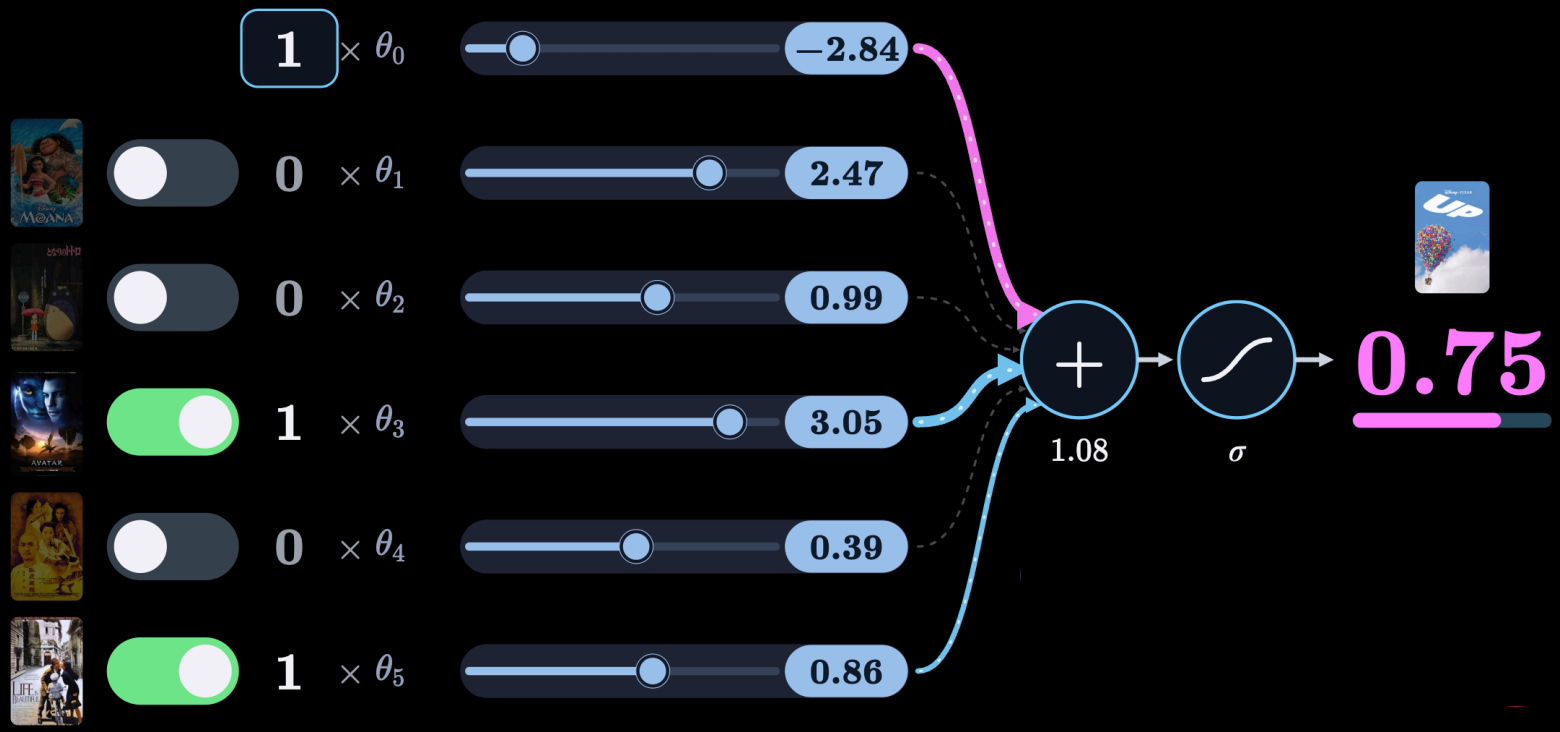




We want the θs that give our 610 viewers the highest log likelihood.

How do we calculate log likelihood?

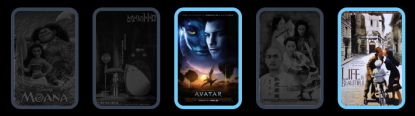
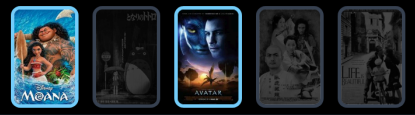
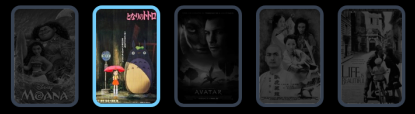
Likelihood



$\mathrm{P}(\text{label is yes} \mid \text{inputs}) = 0.75$

$\mathrm{P}(\text{label is no} \mid \text{inputs}) = 0.25$

Likelihood

from the data		from the model	computed	
viewer	label	$P(\text{watched Up} \mid x)$	$P(\text{label})$ yes: p · no: $1-p$	log
0 0 1 0 1 	yes	0.75	0.75	-0.125
0 0 0 0 0 	no	0.06	0.94	-0.027
1 0 1 0 0 	yes	0.94	0.94	-0.027
0 1 0 0 0 	no	0.14	0.86	-0.066

$$\text{likelihood} = 0.75 \times 0.94 \times 0.94 \times 0.86 \approx 0.57$$

$$\log \text{likelihood} = -0.125 - 0.027 - 0.027 - 0.066 = -0.245$$

A good model: high likelihood. A bad model: low likelihood

Gradient ascent

Nudge every θ a small step uphill, then do it again.

The gradient says which way is uphill for each θ.

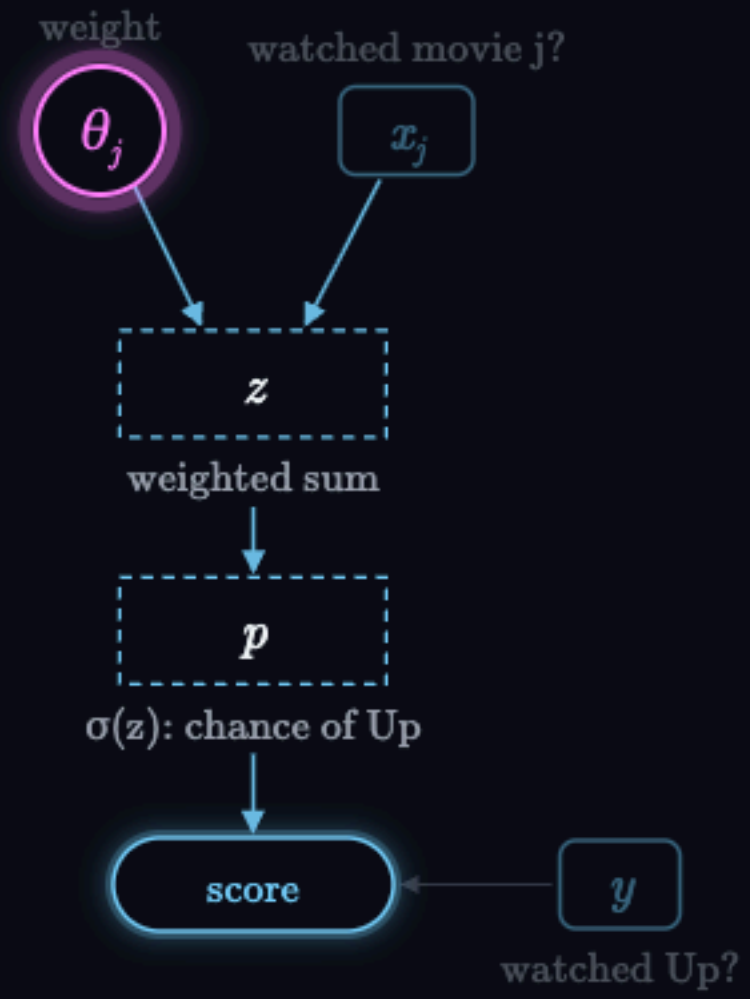
> for the curious

$$\text{new } \theta_j = \theta_j + \text{step size} \times \frac{\partial \, \text{score}}{\partial \theta_j}$$



Code Agent ✓ Done

Find the slope of the score with respect to each θ.



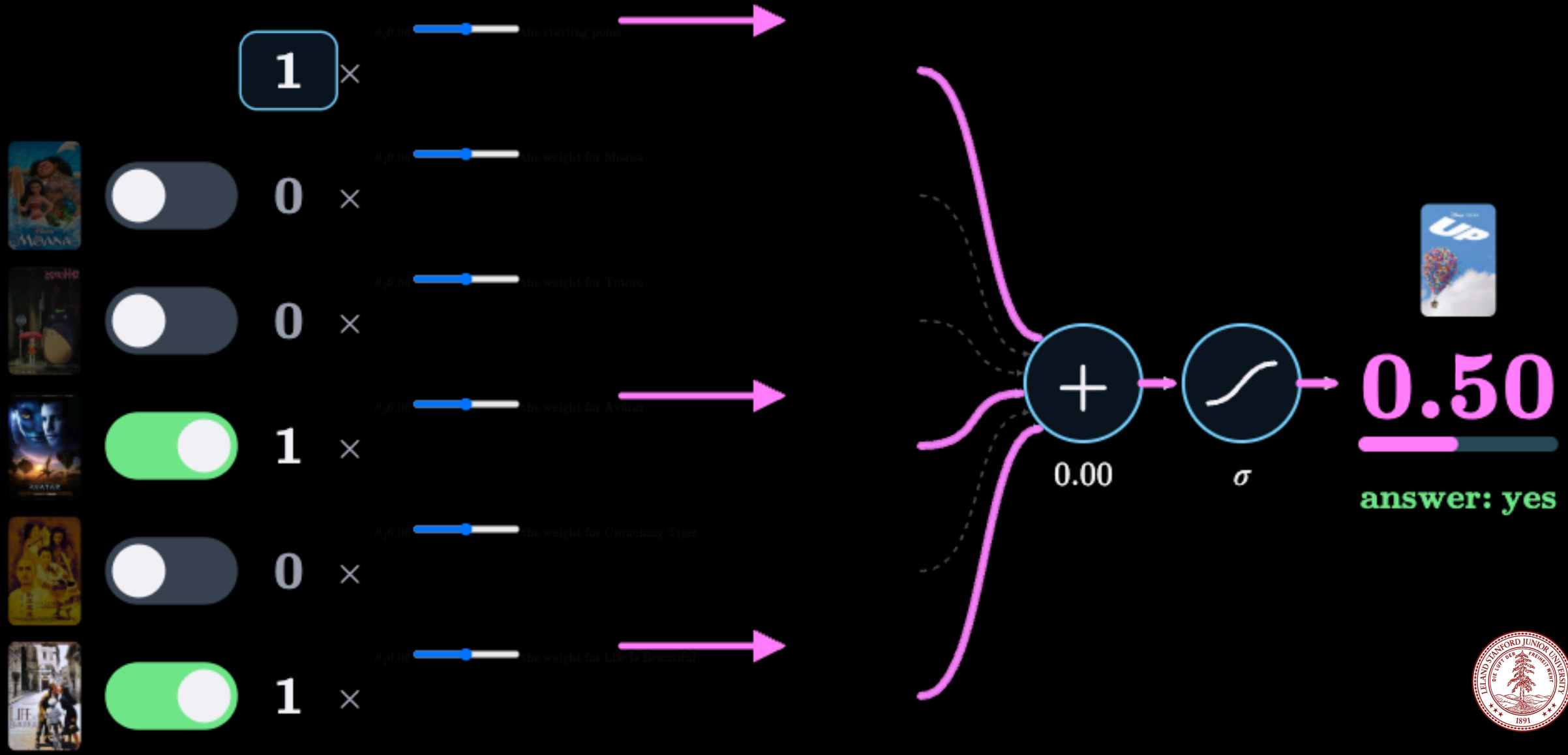
for one example

$$\frac{\partial \text{score}}{\partial \theta_j} = \frac{1}{\ln 10}(y - p)\, x_j$$

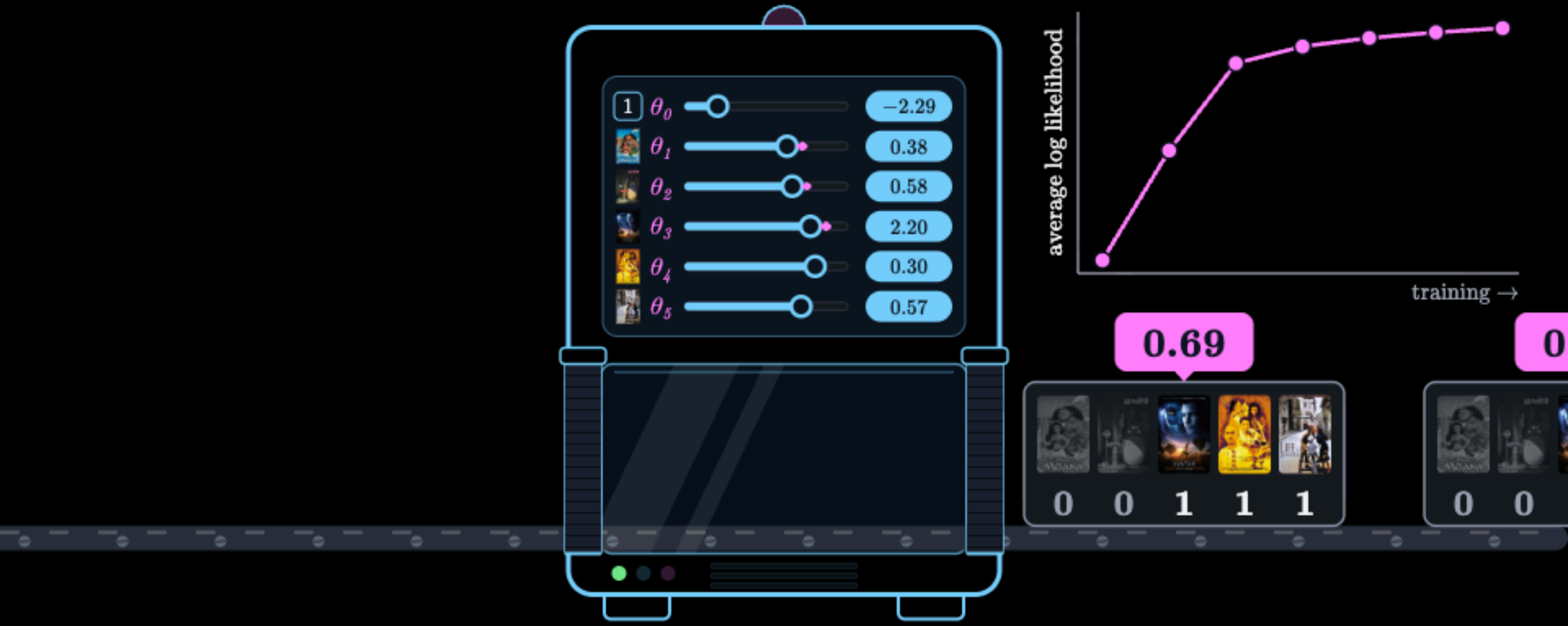
$$\frac{\partial \text{score}}{\partial \theta_j} = \Sigma \frac{1}{\ln 10}(y - \quad p)\, x_j$$

for the whole dataset

Forward and backward passes



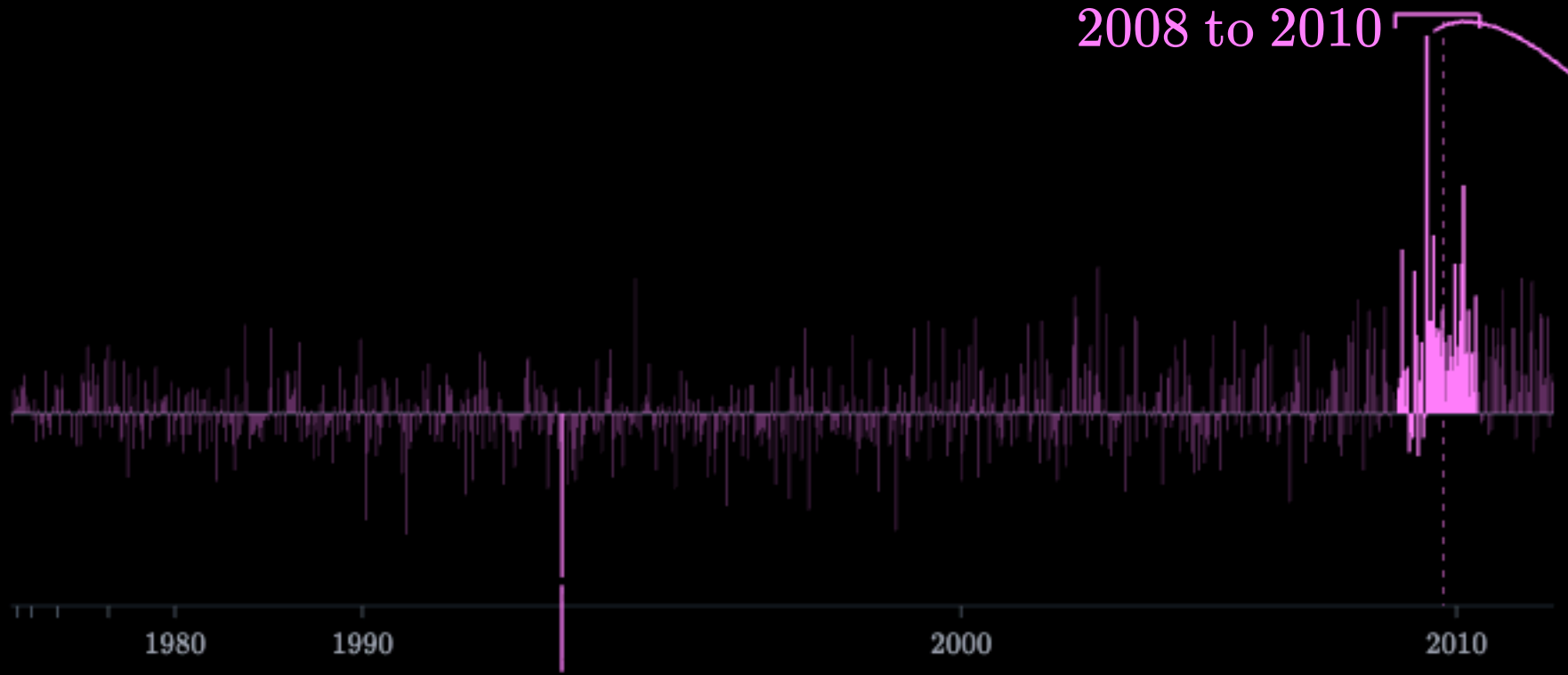
Training


1
θ_0
−2.29
θ_1
0.38
θ_2
0.58
θ_3
2.20
θ_4
0.30
θ_5
0.57
average log likelihood
training →
0.69
0 0 1 1 1
0 0



Training raised the log likelihood from -183.6 to -78.9.

Every movie as a feature





Pulp Fiction, 1994

−0.23

Up, 2009

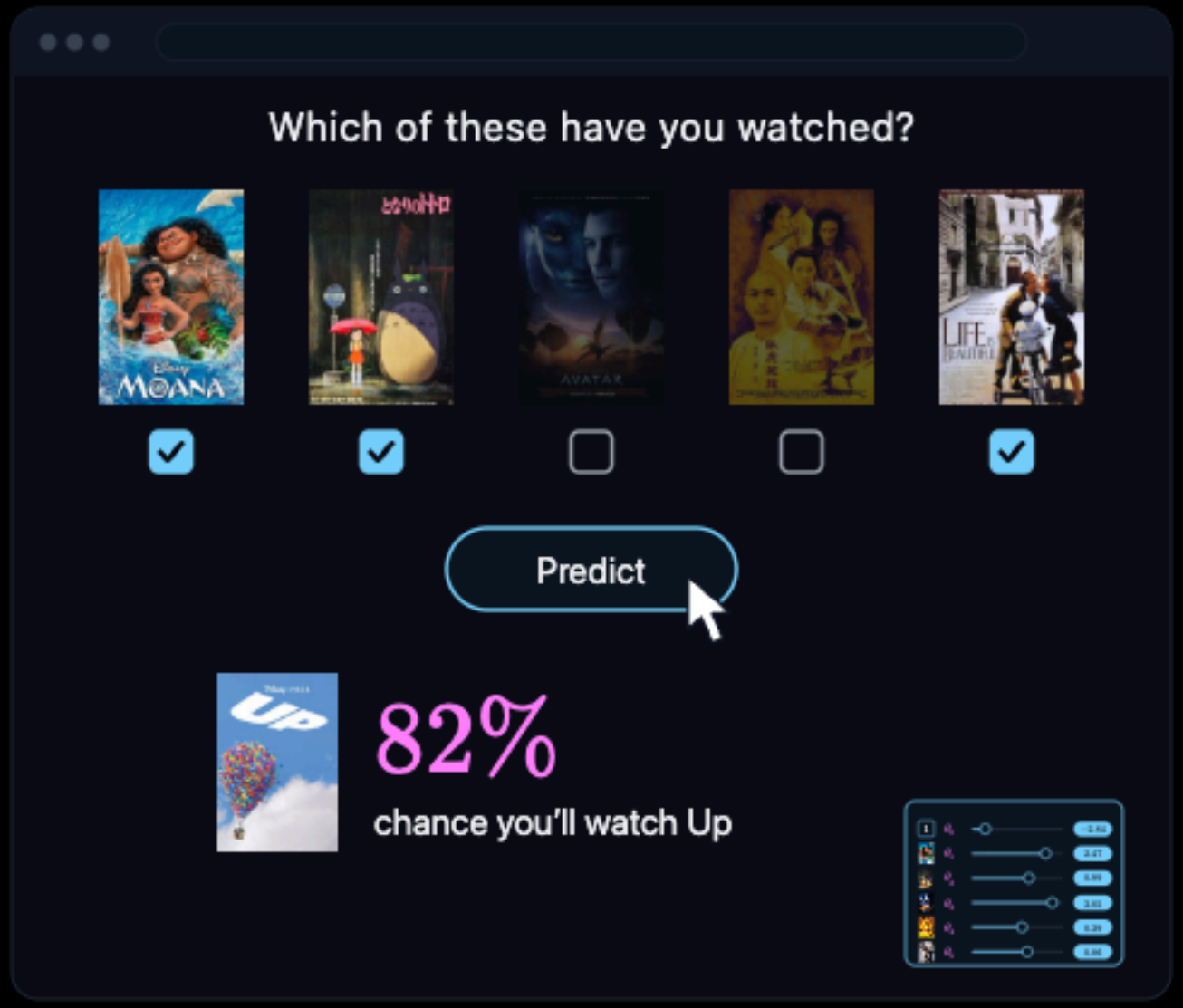


WALL·E

+0.53

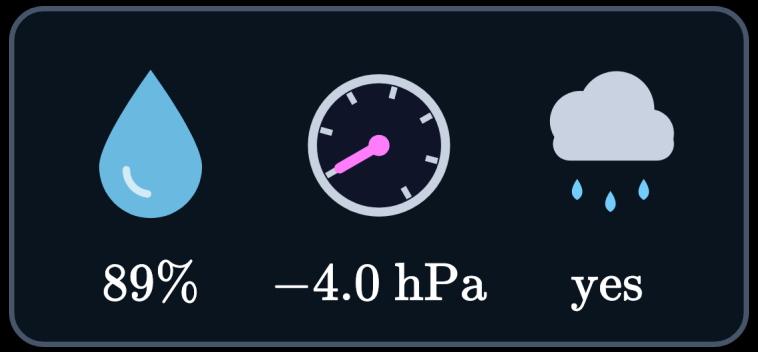
2008

Deploying the model



Deploying the model





Example 2

Rain forecasting

Seattle, May 14, 2023

today



dry

34% humid

tomorrow



rain

3 mm

Probability of precipitation

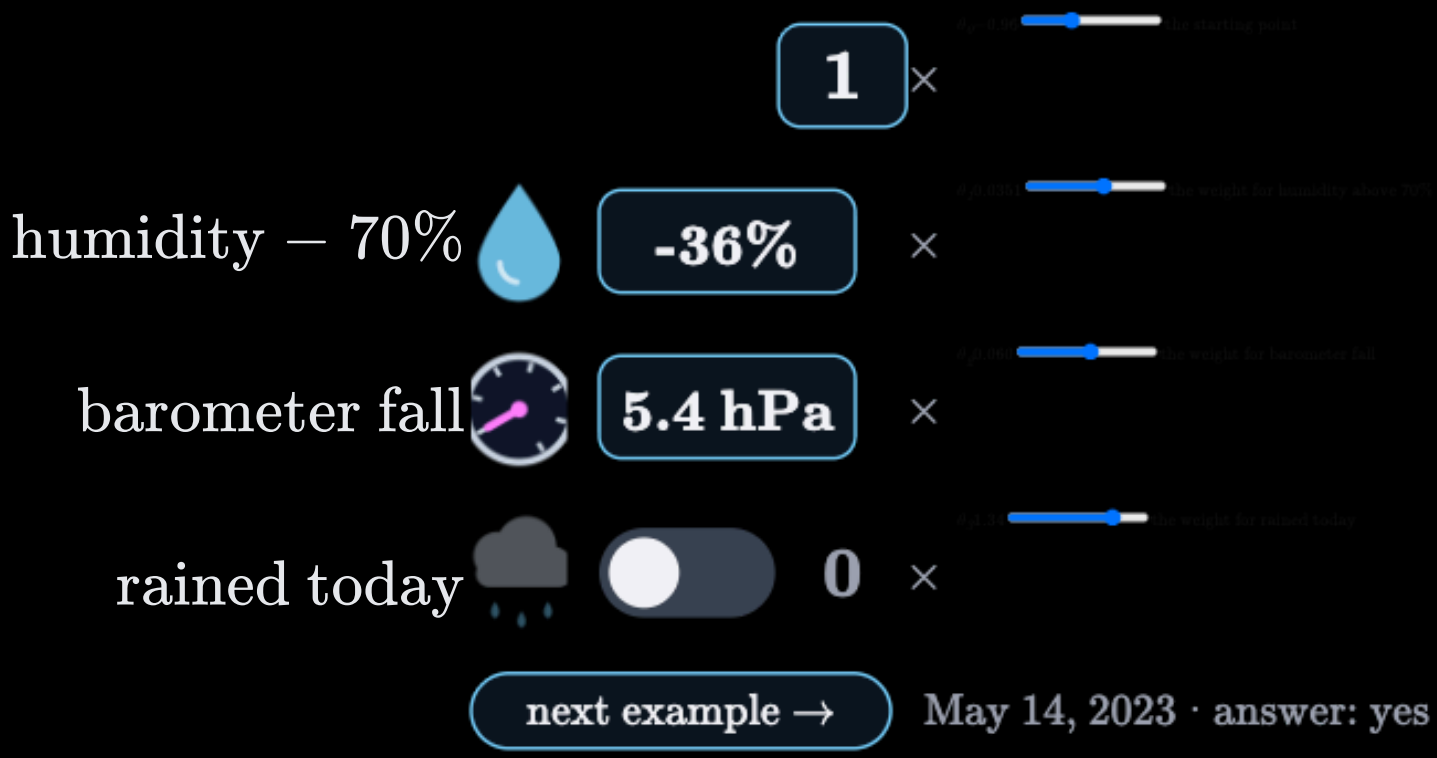


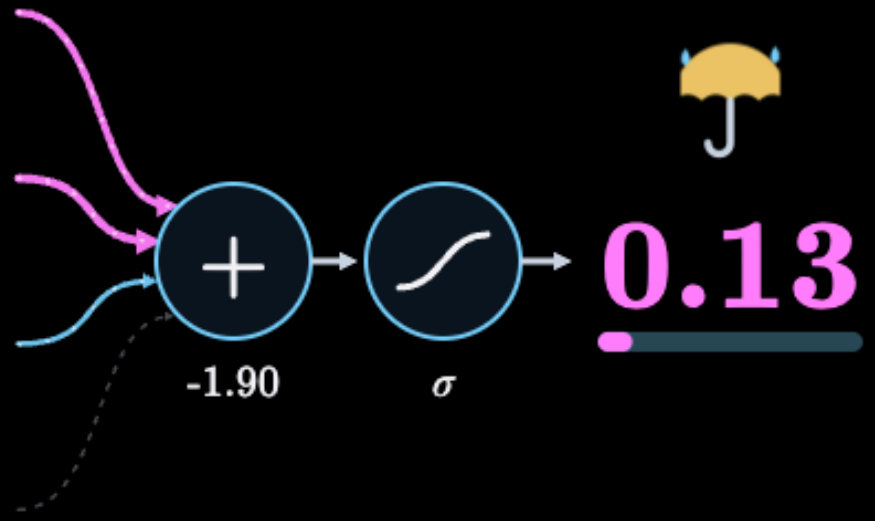
The rain data

	x_1	x_2	x_3	y
				
	humidity	barometer fall	rained today	rain tomorrow?
Nov 1, 2010	89%	−4.0	1	1
Nov 2, 2010	79%	−11.9	1	0
Nov 3, 2010	53%	10.2	0	0
Nov 4, 2010	66%	5.1	0	1
Nov 5, 2010	90%	5.4	1	1

x: what we know y: rain tomorrow?

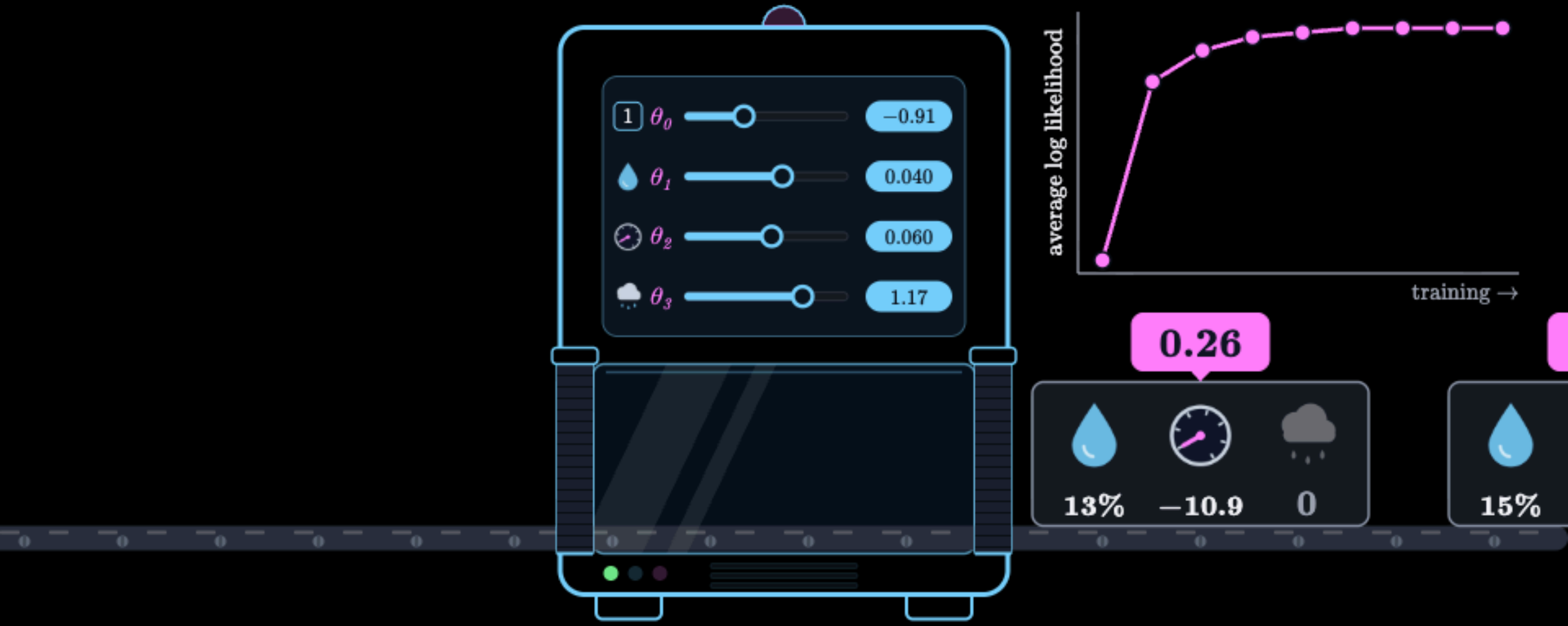
The rain machine





13 in 100. And it rained.

Training the rain machine



Interpreting the weights



humidity $\theta_1 = +0.035$ per 1% 10% more: $+0.35$

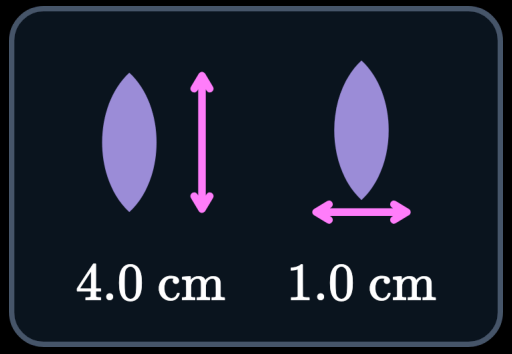


barometer fall $\theta_2 = +0.060$ per hPa 10 hPa rise: -0.60



rained today $\theta_3 = +1.34$ it rained: $+1.34$





Example 3

Flower identification

Classifying irises



Iris versicolor



Iris virginica

Given a flower's petal length and width, is it a virginica?



Photos: Ermell, CC BY-SA 4.0 · James St. John, CC BY 2.0

Code Agent ✓ Done

Show me five flowers going through the trained model.

The trained model



Each number is the model's chance that the flower is a virginica.

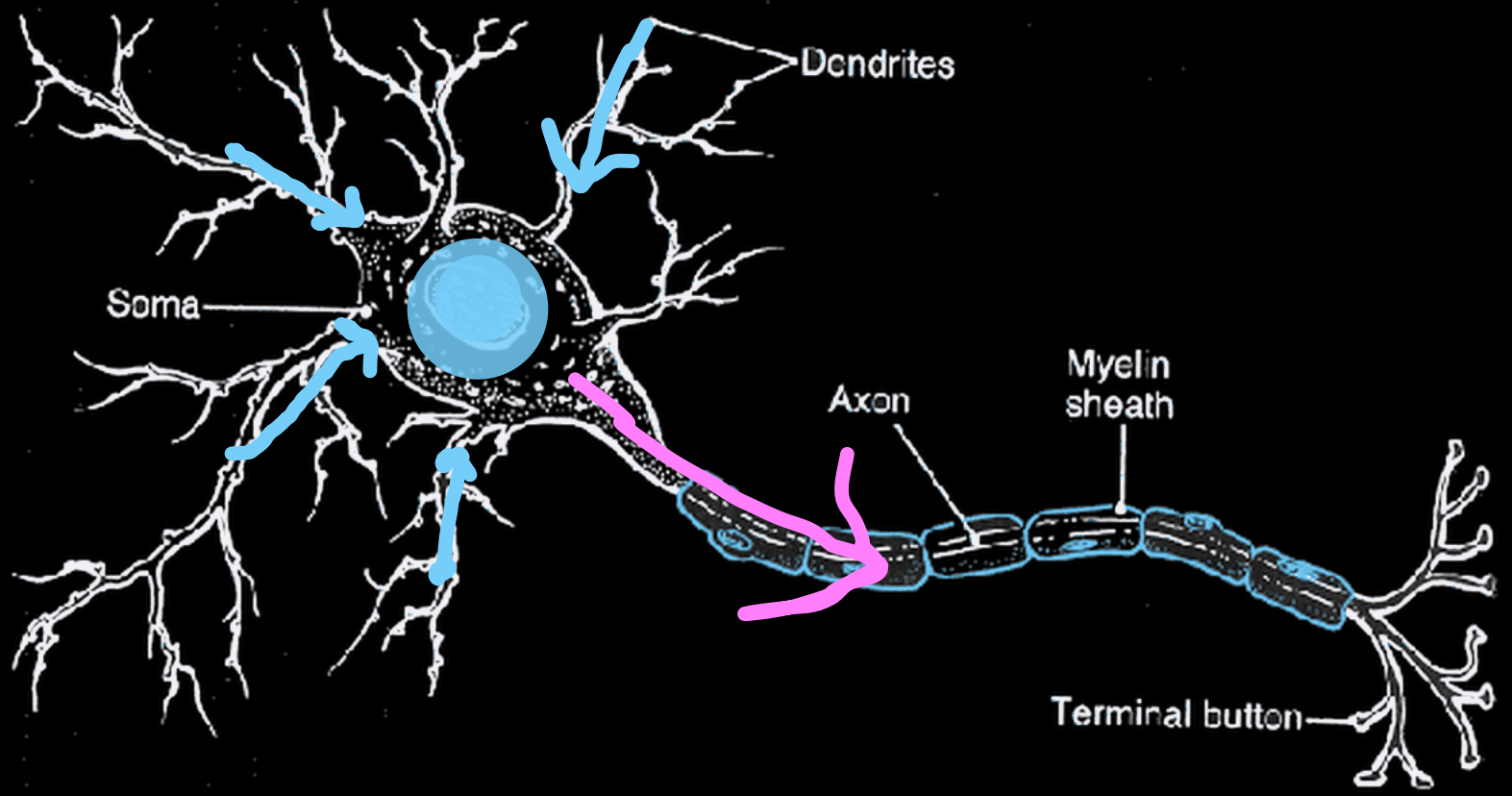
Message Code Agent...



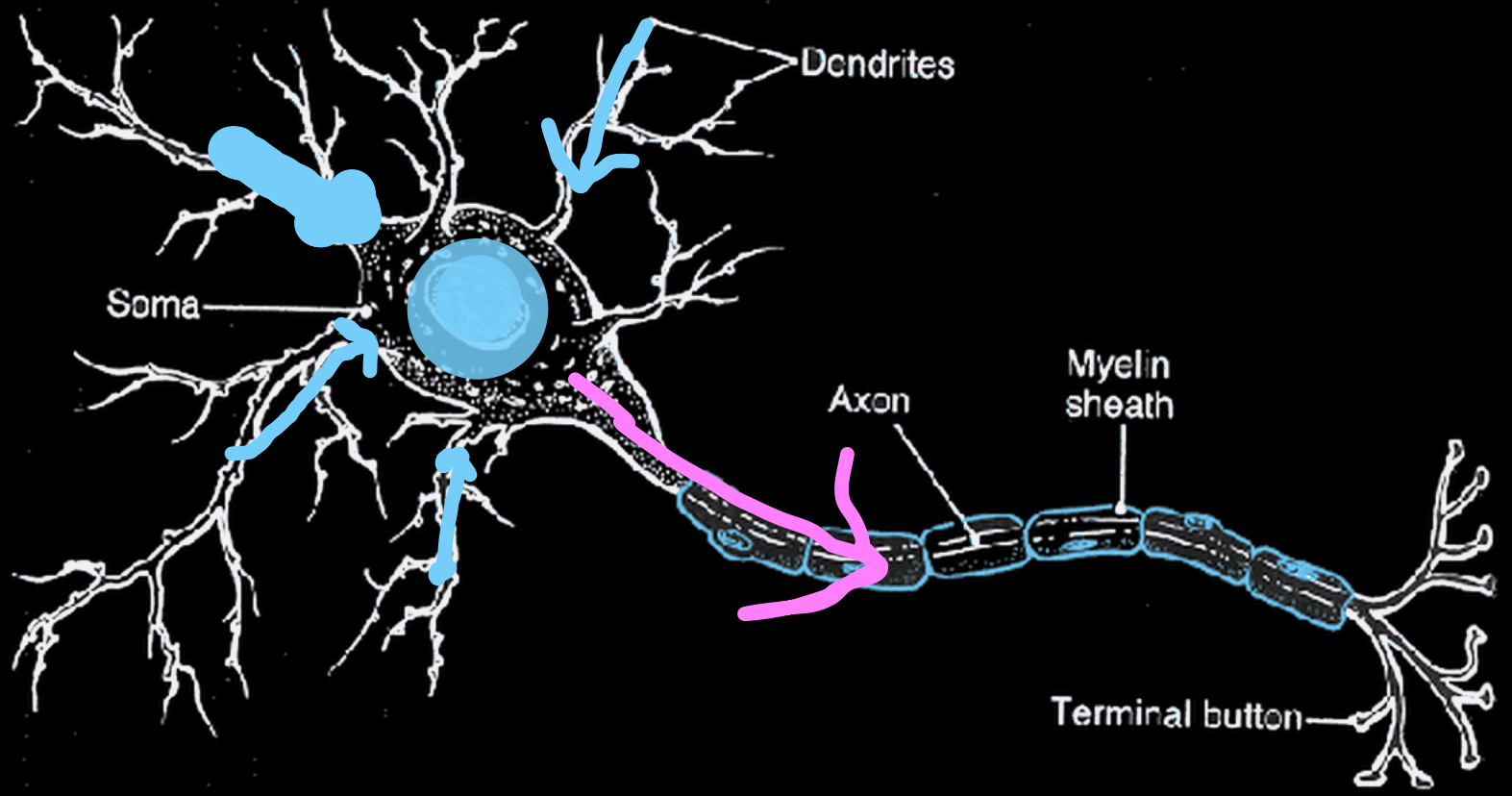

The iris model got 95 of 100 flowers right,
using two petal measurements.

Why is a classifier called regression?

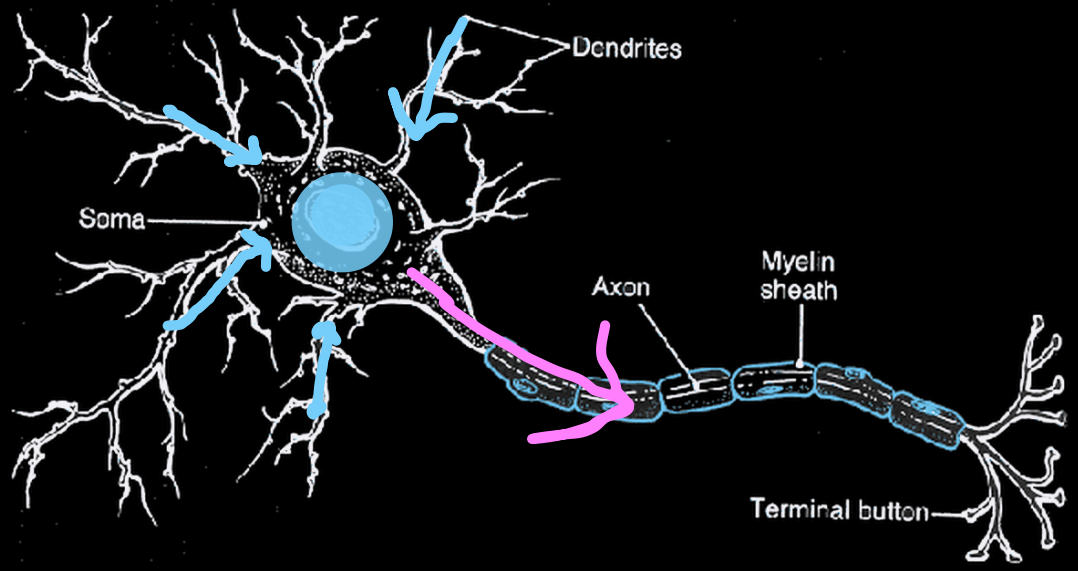
A neuron



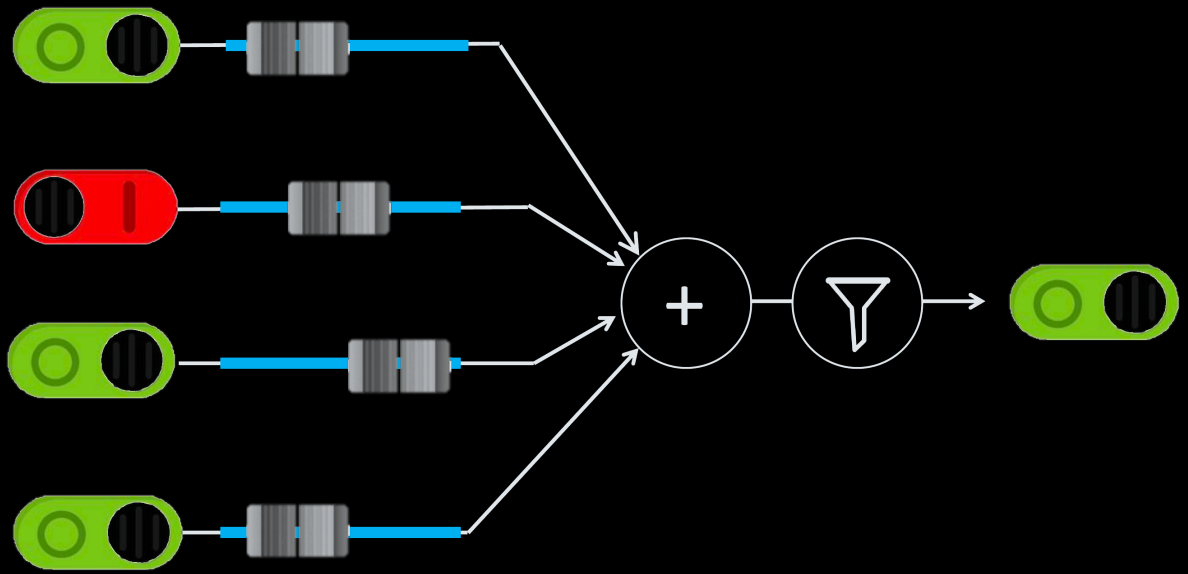
Some inputs are more important



Artificial neurons

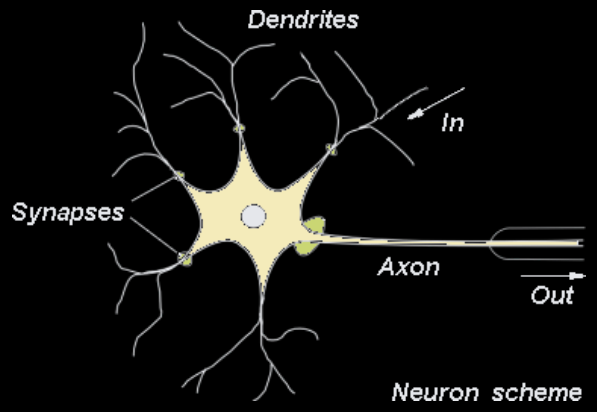


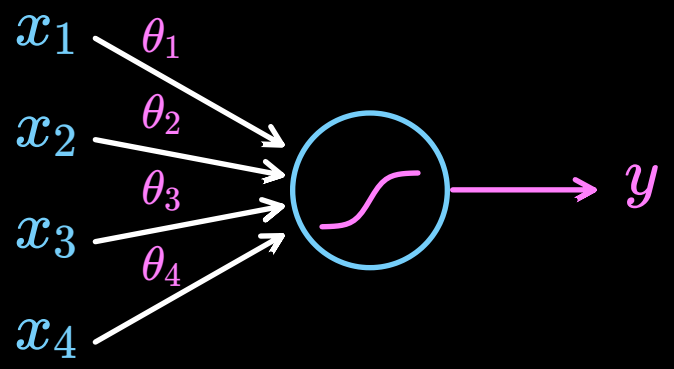




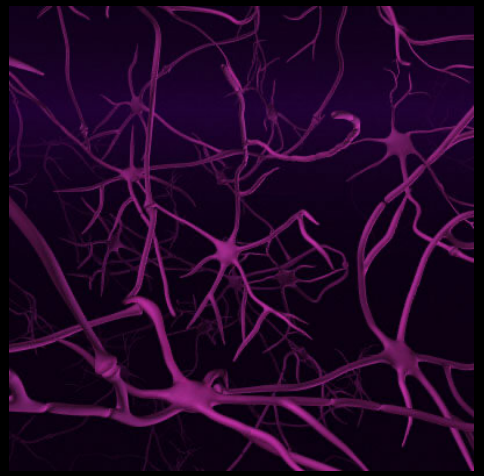
Biological basis for neural networks

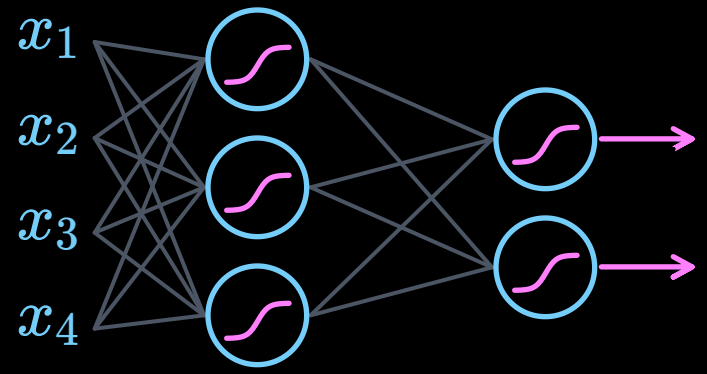
A neuron



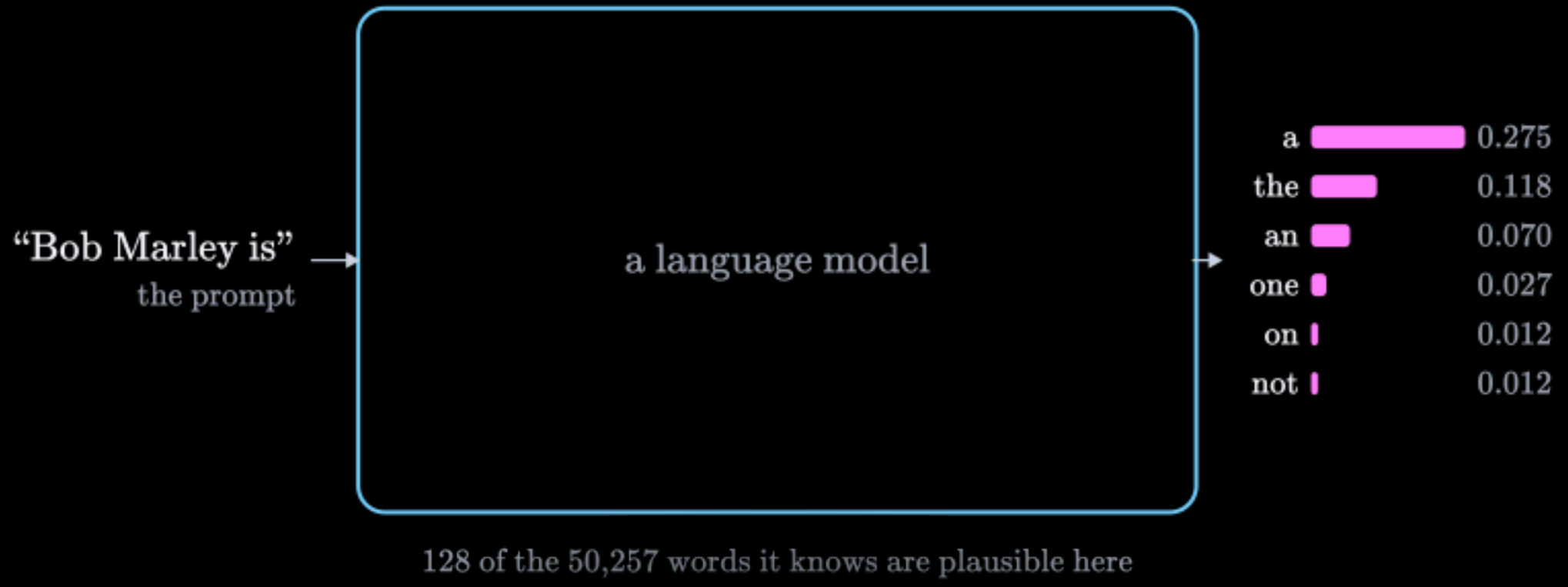


Your brain*





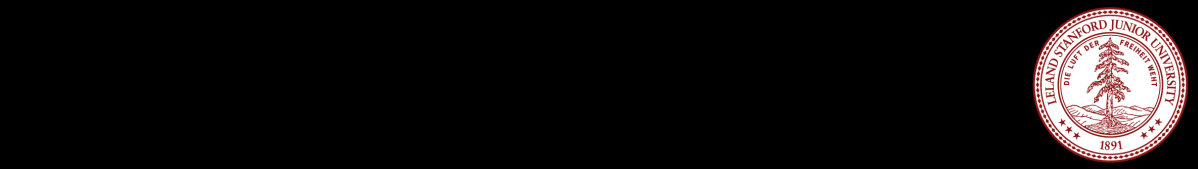
* Actually, it's probably someone else's brain



A language model

Keys

- 🔑 Logistic regression is a probabilistic model.
- 🔑 A computer trains it by climbing the log likelihood.
- 🔑 A weight's sign says which way, and its size says how hard.
- 🔑 The output is a probability: 0.13 is not "no".



That’s all folks!

